Discrete Random Variables

ClipSAT Printable Practice Suite (clipsat.org) • Topic 17.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Expected value (mean) of a discrete random variable**

A random variable X has distribution $P(X=1)=0.2$, $P(X=2)=0.5$, $P(X=3)=0.3$. Find $E(X)$.

$E(X)=\sum xP(X=x)=1(0.2)+2(0.5)+3(0.3)=0.2+1.0+0.9=2.1$.

2 **[Foundational]** • **Target: Probability distributions**

A discrete random variable X has the probability distribution: $P(X=1)=0.2$, $P(X=2)=0.3$, $P(X=3)=k$, $P(X=4)=0.1$. Find k, using the fact that probabilities must sum to 1.

$0.2+0.3+k+0.1=1$, so $k=1-0.6=0.4$.

3 **[Foundational]** • **Target: Probability distributions**

A discrete random variable Y has the probability distribution: $P(Y=0)=0.1$, $P(Y=1)=k$, $P(Y=2)=2k$, $P(Y=3)=0.3$. Find k.

$0.1+k+2k+0.3=1$, so $3k=0.6$, giving $k=0.2$.

4 **[Foundational]** • **Target: Probability distributions**

A discrete random variable Z has the probability distribution: $P(Z=1)=0.25$, $P(Z=2)=0.25$, $P(Z=3)=k$, $P(Z=4)=0.15$, $P(Z=5)=0.15$. Find k.

$0.25+0.25+k+0.15+0.15=1$, so $k=1-0.8=0.2$.

5 **[Foundational]** • **Target: Expected value (mean) of a discrete random variable**

A random variable Y has distribution $P(Y=0)=0.1$, $P(Y=1)=0.4$, $P(Y=2)=0.5$. Find $E(Y)$.

$E(Y)=\sum yP(Y=y)=0(0.1)+1(0.4)+2(0.5)=0+0.4+1.0=1.4$.

6 **[Foundational]** • **Target: Expected value (mean) of a discrete random variable**

A random variable Z has distribution $P(Z=1)=0.3$, $P(Z=2)=0.3$, $P(Z=3)=0.4$. Find $E(Z)$.

$E(Z)=\sum zP(Z=z)=1(0.3)+2(0.3)+3(0.4)=0.3+0.6+1.2=2.1$.

7 **[Foundational]** • **Target: Variance of a discrete random variable**

Using the distribution from the previous question ($E(X)=2.1$), find $\text{Var}(X)$, using $\text{Var}(X)=E(X^2)-[E(X)]^2$.

$E(X^2)=1^2(0.2)+2^2(0.5)+3^2(0.3)=0.2+2.0+2.7=4.9$. $\text{Var}(X)=4.9-(2.1)^2=4.9-4.41=0.49$.

© ClipSAT — clipsat.org | All Rights Reserved

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Variance of a discrete random variable**

Using the distribution $P(Y=0)=0.1$, $P(Y=1)=0.4$, $P(Y=2)=0.5$ ($E(Y)=1.4$), find $\text{Var}(Y)$, using $\text{Var}(Y)=E(Y^2)-[E(Y)]^2$.

$E(Y^2)=0^2(0.1)+1^2(0.4)+2^2(0.5)=0+0.4+2.0=2.4$. $\text{Var}(Y)=2.4-(1.4)^2=2.4-1.96=0.44$.

9 **[Intermediate]** • **Target: Variance of a discrete random variable**

Using the distribution $P(Z=1)=0.3$, $P(Z=2)=0.3$, $P(Z=3)=0.4$ ($E(Z)=2.1$), find $\text{Var}(Z)$, using $\text{Var}(Z)=E(Z^2)-[E(Z)]^2$.

$E(Z^2)=1^2(0.3)+2^2(0.3)+3^2(0.4)=0.3+1.2+3.6=5.1$. $\text{Var}(Z)=5.1-(2.1)^2=5.1-4.41=0.69$.

10 **[Intermediate]** • **Target: Linear transformations of a random variable**

A random variable X has $E(X)=5$ and $\text{Var}(X)=3$. Find $E(2X+4)$ and $\text{Var}(2X+4)$.

$E(2X+4)=2E(X)+4=2(5)+4=14$. $\text{Var}(2X+4)=2^2\text{Var}(X)=4(3)=12$.

11 **[Intermediate]** • **Target: Linear transformations of a random variable**

A random variable X has $E(X)=8$ and $\text{Var}(X)=5$. Find $E(3X-2)$ and $\text{Var}(3X-2)$.

$E(3X-2)=3E(X)-2=3(8)-2=22$. $\text{Var}(3X-2)=3^2\text{Var}(X)=9(5)=45$.

12 **[Intermediate]** • **Target: Linear transformations of a random variable**

A random variable Y has $E(Y)=10$ and $\text{Var}(Y)=4$. Find $E(5-2Y)$ and $\text{Var}(5-2Y)$.

$E(5-2Y)=5-2E(Y)=5-2(10)=-15$. $\text{Var}(5-2Y)=(-2)^2\text{Var}(Y)=4(4)=16$.

13 **[Intermediate]** • **Target: The discrete uniform distribution**

A fair six-sided die is rolled, and X is the number shown. Find $E(X)$ and $\text{Var}(X)$, given that for a discrete uniform distribution on $1, \ldots, n$: $E(X)=\frac{n+1}{2}$ and $\text{Var}(X)=\frac{n^2-1}{12}$.

$n=6$: $E(X)=\frac{7}{2}=3.5$. $\text{Var}(X)=\frac{35}{12}\approx 2.92$.

14 **[Intermediate]** • **Target: The discrete uniform distribution**

A fair eight-sided die (labeled 1 to 8) is rolled, and X is the number shown. Find $E(X)$ and $\text{Var}(X)$, using the same formulas.

$n=8$: $E(X)=\frac{9}{2}=4.5$. $\text{Var}(X)=\frac{63}{12}=5.25$.

© ClipSAT — clipsat.org | All Rights Reserved

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: The discrete uniform distribution** • **Cross-Lesson: 13.01 Newton's Laws**

A fair ten-sided die (labeled 1 to 10) is rolled, and X is the number shown. Find $E(X)$ and $\text{Var}(X)$, using the same formulas.

$n = 10$: $E(X) = \frac{11}{2} = 5.5$. $\text{Var}(X) = \frac{99}{12} = 8.25$.

16 **[Challenging]** • **Target: Constructing a probability distribution from a real scenario** • **Cross-Lesson: 14.01 Energy, Work & Power**

A game involves rolling two fair dice and finding the sum X. Find $P(X = 4)$ and $P(X = 7)$.

There are 36 equally likely outcomes. $X = 4$: $(1, 3), (2, 2), (3, 1)$, so $P(X = 4) = \frac{3}{36} = \frac{1}{12}$. $X = 7$: $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$, so $P(X = 7) = \frac{6}{36} = \frac{1}{6}$.

17 **[Challenging]** • **Target: Constructing a probability distribution from a real scenario** • **Cross-Lesson: 15.01 Representation of Data**

Using the same game of rolling two fair dice and finding the sum X, find $P(X = 9)$ and $P(X = 12)$.

$X = 9$: $(3, 6), (4, 5), (5, 4), (6, 3)$, so $P(X = 9) = \frac{4}{36} = \frac{1}{9}$. $X = 12$: $(6, 6)$, so $P(X = 12) = \frac{1}{36}$.

18 **[Challenging]** • **Target: Constructing a probability distribution from a real scenario** • **Cross-Lesson: 16.01 Permutations & Combinations**

Using the same game of rolling two fair dice and finding the sum X, find $P(X = 5)$ and $P(X = 6)$.

$X = 5$: $(1, 4), (2, 3), (3, 2), (4, 1)$, so $P(X = 5) = \frac{4}{36} = \frac{1}{9}$. $X = 6$: $(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)$, so $P(X = 6) = \frac{5}{36}$.

19 **[Challenging]** • **Target: Cumulative distribution function** • **Cross-Lesson: 18.01 Binomial Distribution**

A random variable X has $P(X = 1) = 0.15$, $P(X = 2) = 0.35$, $P(X = 3) = 0.25$, $P(X = 4) = 0.25$. Find $P(X \leq 2)$ and $P(X > 2)$.

$P(X \leq 2) = P(X = 1) + P(X = 2) = 0.15 + 0.35 = 0.5$. $P(X > 2) = 1 - 0.5 = 0.5$.

20 **[Challenging]** • **Target: Cumulative distribution function** • **Cross-Lesson: 19.01 Normal Distribution**

A random variable X has $P(X = 1) = 0.1$, $P(X = 2) = 0.2$, $P(X = 3) = 0.3$, $P(X = 4) = 0.4$. Find $P(X \leq 3)$ and $P(X \geq 2)$.

$P(X \leq 3) = P(X = 1) + P(X = 2) + P(X = 3) = 0.1 + 0.2 + 0.3 = 0.6$. $P(X \geq 2) = 1 - P(X = 1) = 1 - 0.1 = 0.9$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Probability Distribution Tables & Total Probability

The random variable X has probability distribution $P(X = x) = c(x^2 + 1)$ for $x = 0, 1, 2, 3$. Find the constant c.

a. $\frac{1}{18}$ ✓

b. $\frac{1}{14}$

c. $\frac{1}{15}$

d. $\frac{1}{20}$

For $x = 0 : 1c$; $x = 1 : 2c$; $x = 2 : 5c$; $x = 3 : 10c$. Sum $= (1 + 2 + 5 + 10)c = 18c = 1 \Rightarrow c = 1/18$.

22 [Foundational] • Target: Probability Distribution Tables & Total Probability

A fair die is rolled. Let X be the number of rolls until a 6 appears. What kind of distribution does X follow?

a. **Geometric distribution** ✓

b. Binomial distribution

c. Uniform distribution

d. Poisson distribution

The number of independent Bernoulli trials until the first success is modeled by the Geometric distribution.

23 [Foundational] • Target: Probability Distribution Tables & Total Probability

Given the probability distribution of Y: $P(Y = -1) = 0.3, P(Y = 0) = 0.4, P(Y = 2) = 0.3$. Find $P(Y \geq 0)$.

a. 0.7 ✓

b. 0.4

c. 0.3

d. 1.0

$P(Y \geq 0) = P(Y = 0) + P(Y = 2) = 0.4 + 0.3 = 0.7$.

24 [Foundational] • Target: Expected Value $E(X)$ & Moments

Find the expectation $E(X)$ for the distribution: $P(X = 1) = 0.1, P(X = 2) = 0.3, P(X = 3) = 0.4, P(X = 4) = 0.2$.

a. 2.7 ✓

b. 2.5

c. 3.0

d. 2.8

$E(X) = 1(0.1) + 2(0.3) + 3(0.4) + 4(0.2) = 0.1 + 0.6 + 1.2 + 0.8 = 2.7$.

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Expected Value $E(X)$ & Moments

If $E(X) = 5$, what is $E(3X - 4)$?

a. **11** ✓ **b.** 15

c. 19 **d.** 7

By linearity of expectation: $E(3X - 4) = 3E(X) - 4 = 3(5) - 4 = 15 - 4 = 11$.

26 [Foundational] • Target: Expected Value $E(X)$ & Moments

A game charges 5 dollars to play. A fair coin is tossed: Heads pays 8 dollars, Tails pays nothing. What is the expected net gain for the player?

a. **-1.00** ✓ **b.** +1.00

c. -2.00 **d.** +4.00

Expected payout $= 0.5(8) + 0.5(0) = 4$ dollars. Net gain $= 4 - 5 = -1.00$ dollar.

27 [Foundational] • Target: Variance $\text{Var}(X)$ & Standard Deviation

Given $E(X) = 3$ and $E(X^2) = 13$, find $\text{Var}(X)$.

a. **4** ✓ **b.** 10

c. 2 **d.** 16

$\text{Var}(X) = E(X^2) - [E(X)]^2 = 13 - 3^2 = 13 - 9 = 4$.

28 [Foundational] • Target: Variance $\text{Var}(X)$ & Standard Deviation

If $\text{Var}(X) = 6$, find the variance of $Y = 2X - 5$.

a. **24** ✓ **b.** 12

c. 7 **d.** 36

$\text{Var}(aX + b) = a^2\text{Var}(X)$. Here $\text{Var}(2X - 5) = 2^2 \times 6 = 4 \times 6 = 24$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: Variance $\text{Var}(X)$ & Standard Deviation

Calculate the standard deviation of X given $P(X = 0) = 0.5$ and $P(X = 2) = 0.5$.

a. **1** ✓

b. 2

c. 0.5

d. $\sqrt{2}$

$E(X) = 0(0.5) + 2(0.5) = 1$. $E(X^2) = 0(0.5) + 4(0.5) = 2$. $\text{Var}(X) = 2 - 1^2 = 1$. Standard deviation $\sigma = \sqrt{1} = 1$.

30 [Intermediate] • Target: Multi-Stage Discrete Problem Solving

Two fair 4-sided dice (numbered 1 to 4) are rolled. Let S be the sum of the two numbers. Find $P(S > 5)$.

a. $\frac{6}{16} = \frac{3}{8}$ ✓

b. $\frac{1}{2}$

c. $\frac{5}{16}$

d. $\frac{1}{4}$

Total outcomes $= 16$. Pairs with sum > 5: sum 6 has (2,4), (3,3), (4,2) (3 pairs); sum 7 has (3,4), (4,3) (2 pairs); sum 8 has (4,4) (1 pair). Total favorable $= 3 + 2 + 1 = 6$. Probability $= 6/16 = 3/8$.

31 [Intermediate] • Target: Multi-Stage Discrete Problem Solving

For the sum S of the two 4-sided dice in the previous question, what is $E(S)$?

a. **5** ✓

b. 4

c. 6

d. 4.5

By symmetry, $E(S) = E(D_1) + E(D_2) = 2.5 + 2.5 = 5$.

32 [Intermediate] • Target: Multi-Stage Discrete Problem Solving

A box has 3 red and 2 blue balls. Two balls are drawn without replacement. Let X be the number of blue balls drawn. Find $\text{Var}(X)$.

a. 0.36 ✓

b. 0.40

c. 0.48

d. 0.24

$P(X = 0) = \binom{3}{2}/\binom{5}{2} = 3/10 = 0.3$. $P(X = 1) = (3 \times 2)/10 = 6/10 = 0.6$.
$P(X = 2) = \binom{2}{2}/10 = 1/10 = 0.1$. $E(X) = 0(0.3) + 1(0.6) + 2(0.1) = 0.8$.
$E(X^2) = 0 + 1(0.6) + 4(0.1) = 1.0$. $\text{Var}(X) = 1.0 - 0.8^2 = 1.0 - 0.64 = 0.36$.

© ClipSAT — clipsat.org | All Rights Reserved

33 [Intermediate] • Target: Probability distributions

A discrete random variable, X, can take the values 1, 2, 3, 4, and 5. The probability distribution of X is given by $P(X = x) = kx$ for $x \in \{1, 2, 3, 4, 5\}$, where k is a constant. Find the value of k.

a. $\frac{1}{15}$ ✓

b. $\frac{1}{5}$

c. $\frac{1}{3}$

d. 1

The sum of probabilities for all possible values of a discrete random variable must equal 1. Therefore, $\sum_{x=1}^{5} P(X = x) = 1$. Given $P(X = x) = kx$, we have: $P(X = 1) = k(1) = k$ $P(X = 2) = k(2) = 2k$ $P(X = 3) = k(3) = 3k$ $P(X = 4) = k(4) = 4k$ $P(X = 5) = k(5) = 5k$ Summing these probabilities: $k + 2k + 3k + 4k + 5k = 1$ $15k = 1$ $k = \frac{1}{15}$ Therefore, the value of k is $\frac{1}{15}$.

34 [Intermediate] • Target: Expected value (mean) of a discrete random variable

A discrete random variable X has distribution $P(X = 0) = 1/3, P(X = 1) = 1/2, P(X = 2) = 1/12, P(X = 3) = 1/12$. Find $\text{Var}(X)$.

a. 3/20

b. **107/144** ✓

c. 11/34

d. 1

$E(X) = 11/12, E(X^2) = 19/12$. $\text{Var}(X) = E(X^2) - [E(X)]^2 = 19/12 - (11/12)^2 = 107/144$

35 [Intermediate] • Target: Variance of a discrete random variable

A discrete random variable X has distribution $P(X = 1) = 2/5, P(X = 2) = 1/2, P(X = 3) = 1/10$. Find $\text{Var}(X)$.

a. 3/5

b. 8/9

c. **41/100** ✓

d. 6/7

$E(X) = 17/10, E(X^2) = 33/10$. $\text{Var}(X) = E(X^2) - [E(X)]^2 = 33/10 - (17/10)^2 = 41/100$

36 [Intermediate] • Target: Linear transformations of a random variable

A discrete random variable X has distribution $P(X = 1) = 3/5, P(X = 2) = 1/10, P(X = 3) = 3/10$. Find $\text{Var}(X)$.

a. 7/11

b. 7/10

c. **81/100** ✓

d. 11/21

$E(X) = 17/10, E(X^2) = 37/10$. $\text{Var}(X) = E(X^2) - [E(X)]^2 = 37/10 - (17/10)^2 = 81/100$

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: The discrete uniform distribution

A discrete random variable X has distribution $P(X=1)=4/5, P(X=2)=1/10, P(X=3)=1/10$. Find $\text{Var}(X)$.

a. 1/2

b. **41/100 ✓**

c. 4/5

d. 1/6

$E(X)=13/10, E(X^2)=21/10$. $\text{Var}(X)=E(X^2)-[E(X)]^2=21/10-(13/10)^2=41/100$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Constructing a probability distribution from a real scenario • Cross-Lesson: 16.01 Permutations & Combinations

A discrete random variable X has distribution $P(X=1)=3/10, P(X=2)=1/10, P(X=3)=3/5$. Find $\text{Var}(X)$.

a. 15/28

b. 2/5

c. 25/34

d. **81/100 ✓**

$E(X)=23/10, E(X^2)=61/10$. $\text{Var}(X)=E(X^2)-[E(X)]^2=61/10-(23/10)^2=81/100$

39 [Challenging] • Target: Cumulative distribution function • Cross-Lesson: 18.01 Binomial Distribution

A discrete random variable X has distribution $P(X=1)=2/5, P(X=2)=3/10, P(X=3)=3/10$. Find $\text{Var}(X)$.

a. 12/35

b. **69/100 ✓**

c. 5/12

d. 13/34

$E(X)=19/10, E(X^2)=43/10$. $\text{Var}(X)=E(X^2)-[E(X)]^2=43/10-(19/10)^2=69/100$

© ClipSAT — clipsat.org | All Rights Reserved

40 [Challenging] • Target: A well-designed word problem: a raffle ticket game • Cross-Lesson: 19.01 Normal Distribution

A discrete random variable X has distribution $P(X=1)=1/5, P(X=2)=3/10, P(X=3)=1/2$. Find $\text{Var}(X)$.

a. 61/100 ✓

b. 8/17

c. 6/19

d. 1/9

$E(X)=23/10, E(X^2)=59/10$. $\text{Var}(X)=E(X^2)-[E(X)]^2=59/10-(23/10)^2=61/100$

41 [Challenging] • Target: Probability Distribution Tables & Total Probability • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

A discrete random variable X has distribution $P(X=-3)=7/12, P(X=-1)=1/12, P(X=1)=1/12, P(X=4)=1/4$. Find $\text{Var}(X)$.

a. 425/48 ✓

b. 38/41

c. 11/35

d. 1

$E(X)=-3/4, E(X^2)=113/12$. $\text{Var}(X)=E(X^2)-[E(X)]^2=113/12-(-3/4)^2=425/48$

42 [Challenging] • Target: Expected Value $E(X)$ & Moments • Cross-Lesson: 3.01 Graphs & Transformations

A discrete random variable X has distribution $P(X=0)=1/6, P(X=1)=1/12, P(X=2)=5/12, P(X=3)=1/3$. Find $\text{Var}(X)$.

a. 1

b. 31/35

c. 155/144 ✓

d. 1/16

$E(X)=23/12, E(X^2)=19/4$. $\text{Var}(X)=E(X^2)-[E(X)]^2=19/4-(23/12)^2=155/144$

43 [Challenging] • Target: Variance $\text{Var}(X)$ & Standard Deviation • Cross-Lesson: 4.01 Coordinate Geometry

A discrete random variable X has distribution $P(X=-2)=1/5, P(X=0)=2/5, P(X=3)=1/10, P(X=4)=3/10$. Find $\text{Var}(X)$.

a. 529/100 ✓

b. 40/43

c. 20/41

d. 5/13

$E(X)=11/10, E(X^2)=13/2$. $\text{Var}(X)=E(X^2)-[E(X)]^2=13/2-(11/10)^2=529/100$

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Multi-Stage Discrete Problem Solving • Cross-Lesson: 5.01 Trigonometry

A discrete random variable X has distribution $P(X=-3)=1/10, P(X=0)=3/20, P(X=2)=1/20, P(X=5)=7/10$. Find $\text{Var}(X)$.

a. 771/100 ✓

b. 2/41

c. 25/31

d. 5/7

$E(X)=33/10$, $E(X^2)=93/5$. $\text{Var}(X)=E(X^2)-[E(X)]^2=93/5-(33/10)^2=771/100$

45 [Challenging] • Target: Probability distributions • Cross-Lesson: 6.01 Differentiation & Integration

A discrete random variable X has distribution $P(X=1)=1/2, P(X=2)=3/10, P(X=3)=1/5$. Find $\text{Var}(X)$.

a. 61/100 ✓

b. 7/12

c. 1/19

d. 15/16

$E(X)=17/10$, $E(X^2)=7/2$. $\text{Var}(X)=E(X^2)-[E(X)]^2=7/2-(17/10)^2=61/100$

© ClipSAT — clipsat.org | All Rights Reserved